



11816 - Solving the enigmatic nature of argonium in the Crab Nebula

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI MRS Crab Nebula K1	MIRI Medium Resolution Spectroscopy	(1) Crab-Nebula-K1
	2	MIRI MRS Crab Nebula B2	MIRI Medium Resolution Spectroscopy	(3) Crab-Nebula-B2
	3	MIRI MRS Crab Nebula A1	MIRI Medium Resolution Spectroscopy	(5) Crab-Nebula-A1
	4	MIRI MRS Crab Nebula A2	MIRI Medium Resolution Spectroscopy	(6) Crab-Nebula-A2
	5	MIRI MRS Crab Nebula A12	MIRI Medium Resolution Spectroscopy	(7) Crab-Nebula-A12
	6	MIRI MRS Crab Nebula Background	MIRI Medium Resolution Spectroscopy	(8) Crab-Nebula-Background
	7	NIRSpec Crab Nebula K1	NIRSpec IFU Spectroscopy	(1) Crab-Nebula-K1

Folder	Observation	Label	Observing Template	Science Target
	8	NIRSpec Crab Nebula B1	NIRSpec IFU Spectroscopy	(2) Crab-Nebula-B1
	9	NIRSpec Crab Nebula B2	NIRSpec IFU Spectroscopy	(3) Crab-Nebula-B2
	10	NIRSpec Crab Nebula D4	NIRSpec IFU Spectroscopy	(4) Crab-Nebula-D4
	11	NIRSpec Crab Nebula A1	NIRSpec IFU Spectroscopy	(5) Crab-Nebula-A1
	12	NIRSpec Crab Nebula A2	NIRSpec IFU Spectroscopy	(6) Crab-Nebula-A2
	13	NIRSpec Crab Nebula A12	NIRSpec IFU Spectroscopy	(7) Crab-Nebula-A12
	14	NIRSpec Crab Nebula Background	NIRSpec IFU Spectroscopy	(8) Crab-Nebula-Background

ABSTRACT

For over half a century, the Crab Nebula has served as a critical laboratory for discoveries about supernovae and their remnants. Yet, considerably less attention has been paid to the gas surrounding the synchrotron nebula—an unusually complex and energetic environment. The survival of ArH⁺ emission here was unexpected, as this ion typically traces diffuse, predominantly atomic gas in the ISM. Chemical modeling suggests that ArH⁺ in the Crab originates in denser material exposed to an exceptionally high cosmic-ray ionization rate, raising a key question: does the emission arise from dense H₂-bearing knots or from more diffuse, expanding gas? To address this, we propose high-resolution spectroscopic observations of ArH⁺'s chemical precursors, [Ar II] and H₂, using JWST's MIRI/MRS and NIRSpec instruments. ArH⁺ likely arises from a thin layer where Ar⁺ and warm H₂ coexist under strong cosmic-ray irradiation. The high spatial and spectral resolution's provided by JWST are essential for resolving the filaments and disentangling their distinct kinematics. These measurements will probe a highly energetic and largely unexplored regime of interstellar chemistry. By tracing the origin of ArH⁺ in the Crab, our study will reveal how cosmic rays propagate and interact with multi-phase gas in young remnants, processes that underpin particle-driven chemistry across the Universe. Owing to its proximity, the Crab Nebula also serves as a spatially resolved analogue of the cool-core filaments in galaxy clusters, where similar cosmic-ray-driven processes are thought to shape the thermal and chemical balance of the gas.

OBSERVING DESCRIPTION

The goal of this program is to determine whether the enigmatic ArH⁺ emission in the Crab Nebula originates in the expanding, CR-irradiated outer material or within the dense, partially molecular knots embedded in its filaments. To achieve this, we will trace its chemical precursors-[Ar II] and

H₂-whose emission delineates the formation and destruction pathways of ArH⁺.

We propose JWST spectroscopic observations targeting seven positions in the Crab Nebula with previously confirmed ArH⁺ 617 GHz emission. Two of these positions (labelled here as B1 and D4) were observed with MIRI/MRS in Cycle 1 (but lack NIRSpec observations), while the remaining five will be observed with MIRI/MRS and NIRSpec to complete the sample.

MIRI/MRS observations will cover all four channels and three grating settings, providing full wavelength coverage from 5–28 μ m in 12 sub-bands, encompassing key diagnostic lines including [Ar II] 6.99 μ m, [Ar III] 8.99 μ m, and 21.83 μ m. NIRSpec IFU observations will target in particular H₂ lines demonstrated to be CR-excited, 1–0 S(0), 1–0 Q(2), 1–0 O(2), and 1–0 O(4), using the G235H/F170LP configuration. Each pointing will have a total exposure time of ~1750 s, using a single integration per exposure in the NRSIRS2RAPID readout mode to maximize S/N. Background and leakcal observations will be obtained in identical configurations to ensure accurate background subtraction and correction for possible MSA light leakage.

These observations will yield high spectral-resolution spectra of [Ar II] and H₂ emission, allowing direct comparison with ArH⁺ detections.

The combined dataset will constrain the physical conditions and kinematics of CR-irradiated gas in both expanding filaments and dense knots, providing critical insight into the origin of ArH⁺ and the role of cosmic rays in multi-phase gas chemistry. JWST's unmatched combination of high spatial and spectral resolution is critical for this study, ground-based facilities cannot isolate individual filaments or resolve the velocity substructure required to identify CR-excited H₂ emission and link it to Ar⁺ and subsequently the formation of ArH⁺ formation.

Proposal 11816 - Targets - Solving the enigmatic nature of argonium in the Crab Nebula

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Crab-Nebula-K1	RA: 05 34 34.3100 (83.6429583d) Dec: +21 59 40.60 (21.99461d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(2)	Crab-Nebula-B1	RA: 05 34 34.3400 (83.6430833d) Dec: +21 59 42.10 (21.99503d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(3)	Crab-Nebula-B2	RA: 05 34 32.0400 (83.6335000d) Dec: +21 59 42.00 (21.99500d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(4)	Crab-Nebula-D4	RA: 05 34 29.2800 (83.6220000d) Dec: +22 00 33.50 (22.00931d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(5)	Crab-Nebula-A1	RA: 05 34 33.7500 (83.6406250d) Dec: +21 59 25.10 (21.99031d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(6)	Crab-Nebula-A2	RA: 05 34 31.4500 (83.6310417d) Dec: +21 59 24.90 (21.99025d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(7)	Crab-Nebula-A12	RA: 05 34 32.6000 (83.6358333d) Dec: +21 59 24.70 (21.99019d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular gas, Supernova remnants]</i>	Epoch of Position: 2000	
	(8)	Crab-Nebula-Background	RA: 05 34 38.3095 (83.6596229d) Dec: +21 55 53.53 (21.93154d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Supernova remnants]</i>	Epoch of Position: 2000	

Proposal 11816 - Observation 1 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 1: MIRI MRS Crab Nebula K1													Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI MRS Crab Nebula Background (Obs 6)]													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	Crab-Nebula-K1	RA: 05 34 34.3100 (83.6429583d) Dec: +21 59 40.60 (21.99461d) Equinox: J2000					Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM Description=[Molecular gas, Supernova remnants]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 11816 - Observation 1 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 2 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 2: MIRI MRS Crab Nebula B2													Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI MRS Crab Nebula Background (Obs 6)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(3)	Crab-Nebula-B2	RA: 05 34 32.0400 (83.6335000d)					Epoch of Position: 2000						
			Dec: +21 59 42.00 (21.99500d)											
			Equinox: J2000											
Acquisition	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]													
	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 11816 - Observation 2 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 3 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 3: MIRI MRS Crab Nebula A1													Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI MRS Crab Nebula Background (Obs 6)]													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(5)	Crab-Nebula-A1	RA: 05 34 33.7500 (83.6406250d) Dec: +21 59 25.10 (21.99031d) Equinox: J2000					Epoch of Position: 2000						
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 11816 - Observation 3 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 4 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 4: MIRI MRS Crab Nebula A2												Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI MRS Crab Nebula Background (Obs 6)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(6)	Crab-Nebula-A2	RA: 05 34 31.4500 (83.6310417d) Dec: +21 59 24.90 (21.99025d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction	
		All MRS				NO			FULL			Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 11816 - Observation 4 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 5 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 5: MIRI MRS Crab Nebula A12												Fri Mar 13 22:02:54 GMT 2026		
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
	Background Observations:[MIRI MRS Crab Nebula Background (Obs 6)]														
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name		Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(7)	Crab-Nebula-A12		RA: 05 34 32.6000 (83.6358333d)				Epoch of Position: 2000							
				Dec: +21 59 24.70 (21.99019d)											
				Equinox: J2000											
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]														
Acquisition	#	Target													
	1	NONE													
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			Grating Wheel Direction		
		All MRS				NO				FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction					
	1	4-Point				EXTENDED SOURCE				NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008			
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008			
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008			
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008			
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008			
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008			

Proposal 11816 - Observation 5 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 6 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 6: MIRI MRS Crab Nebula Background												Fri Mar 13 22:02:54 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Background Observation For: [MIRI MRS Crab Nebula K1 (Obs 1), MIRI MRS Crab Nebula B2 (Obs 2), MIRI MRS Crab Nebula A1 (Obs 3), MIRI MRS Crab Nebula A2 (Obs 4), MIRI MRS Crab Nebula A12 (Obs 5)]														
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(8)	Crab-Nebula-Background	RA: 05 34 38.3095 (83.6596229d) Dec: +21 55 53.53 (21.93154d) Equinox: J2000					Epoch of Position: 2000						
	Comments: Category=ISM Description=[Supernova remnants]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					POINT SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 11816 - Observation 6 - Solving the enigmatic nature of argonium in the Crab Nebula

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, 6, Non-interruptible
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Proposal 11816 - Observation 7 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 7: NIRSpec Crab Nebula K1											Fri Mar 13 22:02:54 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	Crab-Nebula-K1		RA: 05 34 34.3100 (83.6429583d) Dec: +21 59 40.60 (21.99461d) Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=ISM Description=[Molecular gas, Supernova remnants]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			SMALL		1		4				
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP		NRSIRS2RAPID	29	1	false	true	NONE	4	4	1750.667	
	2	G235H/F170LP		NRSIRS2RAPID	29	1	true	true	NONE	4	4	1750.667	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible												

Proposal 11816 - Observation 8 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 8: NIRSpec Crab Nebula B1											Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	Crab-Nebula-B1		RA: 05 34 34.3400 (83.6430833d) Dec: +21 59 42.10 (21.99503d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		4			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP	NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible											

Proposal 11816 - Observation 9 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 9: NIRSpec Crab Nebula B2											Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	Crab-Nebula-B2		RA: 05 34 32.0400 (83.6335000d) Dec: +21 59 42.00 (21.99500d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category= <i>ISM</i> Description= <i>[Molecular gas, Supernova remnants]</i>											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP	NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible											

Proposal 11816 - Observation 10 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 10: NIRSpec Crab Nebula D4											Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	Crab-Nebula-D4		RA: 05 34 29.2800 (83.6220000d) Dec: +22 00 33.50 (22.00931d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		4			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP	NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible											

Proposal 11816 - Observation 11 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 11: NIRSpec Crab Nebula A1											Fri Mar 13 22:02:54 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	Crab-Nebula-A1		RA: 05 34 33.7500 (83.6406250d) Dec: +21 59 25.10 (21.99031d) Equinox: J2000			Epoch of Position: 2000						
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			SMALL		1		4				
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP		NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP		NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible												

Proposal 11816 - Observation 12 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 12: NIRSpec Crab Nebula A2											Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]											
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	Crab-Nebula-A2		RA: 05 34 31.4500 (83.6310417d) Dec: +21 59 24.90 (21.99025d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		4			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP	NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible											

Proposal 11816 - Observation 13 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 13: NIRSpec Crab Nebula A12											Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec Crab Nebula Background (Obs 14)]											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	Crab-Nebula-A12		RA: 05 34 32.6000 (83.6358333d) Dec: +21 59 24.70 (21.99019d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=ISM Description=[Molecular gas, Supernova remnants]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		4			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	
	2	G235H/F170LP	NRSIRS2RAPID	15	1	true	true	NONE	4	4	933.689	
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible											

Proposal 11816 - Observation 14 - Solving the enigmatic nature of argonium in the Crab Nebula

Observation	Proposal 11816, Observation 14: NIRSpec Crab Nebula Background												Fri Mar 13 22:02:54 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Background Observation For: [NIRSpec Crab Nebula K1 (Obs 7), NIRSpec Crab Nebula B1 (Obs 8), NIRSpec Crab Nebula B2 (Obs 9), NIRSpec Crab Nebula D4 (Obs 10), NIRSpec Crab Nebula A1 (Obs 11), NIRSpec Crab Nebula A2 (Obs 12), NIRSpec Crab Nebula A12 (Obs 13)]												
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(8)	Crab-Nebula-Background		RA: 05 34 38.3095 (83.6596229d) Dec: +21 55 53.53 (21.93154d) Equinox: J2000			Epoch of Position: 2000						
	Comments: Category= <i>ISM</i> Description= <i>[Supernova remnants]</i>												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-NOD											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G235H/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911		
	2	G235H/F170LP	NRSIRS2RAPID	10	1	true	true	NONE	4	4	641.911		
Special Requirements	Sequence Observations 7, 8, 9, 10, 11, 12, 13, 14, Non-interruptible												